

SAR Despeckling via Fourier Refinement and Representation-Consistent Reconstruction

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Abstract—The discrepancy between simulated speckle and real synthetic aperture radar (SAR) observations limits the generalization of supervised despeckling models. We extend the public Trans-SAR backbone by incorporating full-spectrum Fourier-domain refinement and representation-consistent reconstruction. Specifically, one bottleneck block and five decoder blocks combine local features with real-imaginary channel mixing, and bounded log-intensity predictions are inverted before intensity compensation. Masked adaptation updates the reconstruction on noisy real SAR images while the Transformer encoder remains frozen. Across 8,400 simulated pairs from 21 UCMerced classes and four look numbers, the proposed model achieves macro-averaged peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) values of 28.14 dB and 0.7791, respectively, without requiring the number of looks as input. PSNR is highest among evaluated methods at every look. Removing the Fourier refinement reduces PSNR by 1.68 dB. On 592 patches from held-out parent images, adaptation lowers the M-index from 1.8213 to 1.3365 and improves the edge-preservation index from 0.4229 to 0.8214. These results validate the proposed reconstruction design and adaptation strategy on the evaluated datasets.

Index Terms—Despeckling, Fourier refinement, self-supervised adaptation, synthetic aperture radar, Transformer.

I. INTRODUCTION

In synthetic aperture radar (SAR), speckle obscures edges, texture, and bright scatterers. Despeckling must restore them despite signal-dependent noise and scarce clean references [1]. Unit-mean Gamma corruption enables controlled multilook tests, but focused SAR images also contain spatially correlated speckle and processing artifacts; models trained on simulated pairs therefore often transfer poorly to real observations.

Lee filtering uses local statistics [2]; SAR-BM3D uses nonlocal recurrence and transform-domain shrinkage [3]. Supervised CNNs learn paired priors [4], and SAR-CAM adds continuous multiscale attention [5]. Trans-SAR couples a hierarchical Transformer encoder with convolutional reconstruction [6]; later models add dynamic gates and noise-guided multiscale fusion [7], [8]. SAR-FDD and DATNet use frequency decomposition alongside spatial processing [7], [9], while transformed conditional diffusion addresses multiplicative statistics [10]. Spectral overlap between speckle and scene structure motivates joint local and full-spectrum refinement without predefined bands.

Learning from real observations depends on supervision. SAR2SAR uses multitemporal acquisitions [11]; Speckle2Void combines blind-spot likelihood training and decorrelation [12];

Speckle2Self uses masked self-supervision with attention [13]. SDS-SAR forms intensity-only pairs through correlation-aware sampling [14], whereas SDUDNet uses unpaired clean and noisy images [15]. These methods motivate adaptation without clean targets, but differing assumptions complicate outcome comparisons.

We extend public Trans-SAR in three ways. Six Fourier-domain refinement (FDR) blocks combine local filtering with full-spectrum real-imaginary channel mixing at the bottleneck and five decoder scales. An exact inverse maps bounded log predictions to intensity before residual compensation, keeping fusion in one representation. Encoder-frozen masked post-adaptation (AMS) calibrates reconstruction on noisy real SAR without clean targets. Fig. 1 separates inherited operators from additions. Ablations assess reconstruction components; one model covers four look numbers and a real-SAR split with disjoint parent images. Real-data evidence remains empirical because masked noisy targets cannot guarantee unbiased restoration under correlated speckle.

II. PROPOSED METHOD

A. Observation and Reconstruction Path

Let $\mathbf{X}, \mathbf{Y} \in [0, 1]^{H \times W}$ denote a clean intensity proxy and its noisy observation. Synthetic pairs use independent Gamma speckle [6] with look number L . We clip the simulated intensities to $[0, 1]$:

$$\begin{aligned} \mathbf{Y} &= \text{clip}(\mathbf{X} \odot \mathbf{N}, 0, 1), \\ N_{ij} &\sim \text{Gamma}(L, L^{-1}). \end{aligned} \quad (1)$$

The shape-scale parameterization gives unit mean and variance L^{-1} before clipping. This controlled corruption does not reproduce the full statistics of focused SAR images. We use the normalized logarithmic mapping in (2) and its algebraic inverse in (3):

$$T_\alpha(\mathbf{Y}) = \frac{\log(1 + \alpha \mathbf{Y})}{\log(1 + \alpha)}, \quad (2)$$

$$T_\alpha^{-1}(\hat{\mathbf{Z}}) = \frac{\exp[\hat{\mathbf{Z}} \log(1 + \alpha)] - 1}{\alpha}. \quad (3)$$

The mapping moderates bright intensities without assuming additive Gaussian noise in the transformed domain.

Trans-SAR supplies overlap embeddings, spatial-reduction attention, and basic upsampling [6]. Its five encoder stages have widths (32, 64, 128, 320, 512), head counts (1, 1, 2, 4, 8), depths (3, 3, 4, 6, 3), and resolutions from 128^2 to 8^2 . After bottleneck refinement, five decoder stages apply upsampling, available skip fusion, FDR, residual convolution, and guidance

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Outputs are intensity, and mask ones identify masked pixels. Only the Transformer encoder is frozen: bottleneck, decoder, guidance, head, and compensator adapt for eight epochs at learning rate 10^{-6} . Training masks are resampled; fixed validation masks enable checkpoint selection at identical locations. Inference uses unmasked observations. During AMS, $\mathbf{Y}_m$ replaces $\mathbf{Y}$ throughout the network, including the compensator.

III. EXPERIMENTS

A. Datasets and Evaluation Protocol

NWPU-RESISC45 [17] supplies 900 training and 225 validation images (20/5 per class), with squared grayscale values as clean intensity proxies. A deterministic schedule balances 100,000 training presentations across $L \in \{1, 2, 4, 8\}$; four fixed realizations per validation source yield 900 pairs. Paired flips and right-angle rotations augment training.

All 2,100 UCMerced images from 21 classes [18] are excluded from model selection and yield 8,400 fixed test pairs, one corruption per source at each of four looks. If needed, images are bilinearly resized to 256×256 before corruption by (1), with no subsequent corruption. This tests transfer between optical collections under simulated speckle.

For real SAR, we use only noisy images from Mendeley dataset fs455tz88y, version 1 [19]. The split assigns 1,175 training, 146 validation, and 148 test parent images, yielding 4,700, 584, and 592 patches, respectively. All four children of each 512×512 parent remain together. Reference images are excluded from adaptation, selection, measurement, and visualization.

Methods share test IDs. Unsigned 8-bit inputs are divided by 255; normalized floating-point inputs remain unchanged. Evaluation uses normalized linear intensity without percentile normalization or display stretching.

B. Implementation, Baselines, and Metrics

We train with Adam for 100,000 updates with batch size 1, initial learning rate 10^{-3} , and weight decay 10^{-5} . We tested $\alpha \in \{1, 5, 10, 20, 50\}$ and fixed $\alpha = 10$ by synthetic validation PSNR, supported by SSIM. Before testing, validation sweeps selected $\lambda_{TV}^{\text{sup}} = 3 \times 10^{-2}$ from $\{10^{-2}, 3 \times 10^{-2}, 5 \times 10^{-2}, 8 \times 10^{-2}, 10^{-1}\}$ for the highest synthetic macro PSNR/SSIM, and $\lambda_{\text{AMS}} = 10^{-3}$ from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ for the best real-SAR M-index/EPI tradeoff (ENL diagnostic only); both remain fixed. Every 5,000 updates, the lowest validation

intensity mean squared error (MSE) averaged equally by class and look selects the checkpoint. A plateau scheduler halves the rate after four validation events without improvement, with a 10^{-6} floor. One checkpoint handles all looks. Full and Ours-base denote the pre-adaptation model. NWPU validation selects Lee-MMSE's window; SAR-BM3D uses square-root input and squared output. Both receive nominal L (oracle- L); SAR-CAM [5], Trans-SAR [6], and our model receive only corrupted intensity.

UCM-21 PSNR and SSIM average per-image scores equally across classes and looks. For the 592 real patches, each noisy input yields five nonoverlapping 32×32 regions of interest (ROIs) on a grid, ranked by increasing squared coefficient of variation after rejecting dark, zero-dominated, and saturated candidates. The equivalent number of looks (ENL) averages μ^2/σ^2 over these 2,960 method-shared ROIs.

Both real-SAR evaluations use the M-index [20] and edge-preservation index (EPI) [21]. The former combines ratio-image mean/ENL deviations with homogeneity changes under random permutation; here, EPI is the ratio of summed absolute diagonal-neighbor differences in the output to those in the noisy input. ENL is reported as a mean and diagnoses smoothness rather than overall quality. A lower M-index is preferred; EPI requires joint interpretation with ENL and ratio structure, because residual speckle also increases it [22].

C. Synthetic Comparison

Our single blind model achieves the highest PSNR point estimates in Table I, surpassing the strongest alternatives by 0.40, 0.72, 0.43, and 0.35 dB at $L = 1, 2, 4, 8$, respectively. The macro PSNR is 28.14 dB, 0.48 dB above Trans-SAR and 0.66 dB above oracle- L SAR-BM3D. The macro SSIM is 0.7791, with respective gains of 0.0755 and 0.0027. SAR-BM3D leads SSIM at $L = 1, 4$, whereas ours is higher at $L = 2, 8$.

In Fig. 2, our restoration suppresses speckle while retaining roof contours and rooftop structures, whereas LEE blurs these details. The building outlines remain in our ratio image, indicating incomplete separation of structure from speckle.

D. Generalization across Three UCMerced Scene Types

Table II separately evaluates 100 agricultural, 100 buildings, and 100 dense residential UCMerced samples at $L = 4$, representing smooth cover, regular boundaries, and dense texture. These sources are excluded from supervised development.

TABLE I
UCM-21 GAMMA-SPECKLE RESULTS ON 8,400 FIXED PAIRS

Method	Look mode	$L = 1$	$L = 2$	$L = 4$	$L = 8$	Macro
		PSNR / SSIM	PSNR / SSIM	PSNR / SSIM	PSNR / SSIM	PSNR / SSIM
Noisy	–	12.95 / 0.2069	14.98 / 0.2870	17.37 / 0.3840	20.01 / 0.4916	16.33 / 0.3424
Lee-MMSE	oracle- L	21.23 / 0.5212	23.05 / 0.5999	24.71 / 0.6757	26.33 / 0.7455	23.83 / 0.6356
SAR-BM3D	oracle- L	25.24 / 0.7234	26.78 / <u>0.7612</u>	28.14 / 0.7967	29.77 / <u>0.8243</u>	27.48 / <u>0.7764</u>
SAR-CAM	blind	22.68 / 0.5622	23.54 / 0.6138	25.89 / 0.7295	28.10 / 0.7949	25.05 / 0.6751
Trans-SAR	blind	<u>25.53</u> / 0.6446	<u>27.22</u> / 0.6963	28.12 / 0.7397	<u>29.78</u> / 0.7336	<u>27.66</u> / 0.7036
Ours (Full)	blind	25.93 / <u>0.7169</u>	27.94 / 0.7930	28.57 / <u>0.7754</u>	30.13 / 0.8312	28.14 / 0.7791

PSNR is in dB. Best and second-best values among despeckling methods are bold and underlined, respectively. Oracle- L methods receive the nominal look number; blind methods do not.

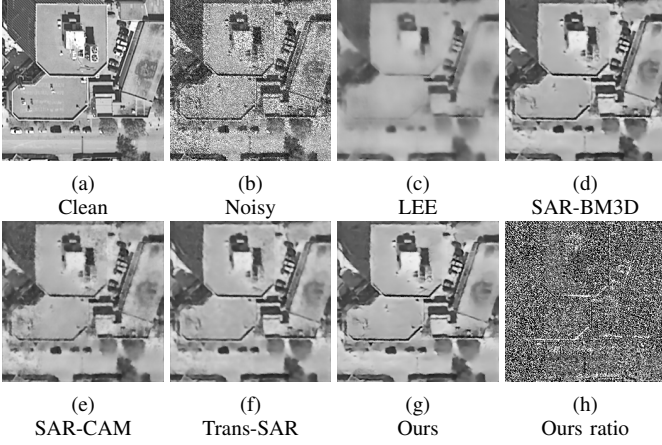


Fig. 2. Visual comparison for a UCMerced buildings scene. LEE denotes Lee-MMSE. All panels show the same field of view in their original grayscale renderings. Panel (h) shows the ratio image for Ours and is assessed separately from the restorations.

TABLE II
GENERALIZATION ON THREE UCMERCE SCENE TYPES AT $L = 4$

Scene	Num.	Noisy	Ours-base
		PSNR / SSIM	PSNR / SSIM
Agricultural	100	17.7573 / 0.4218	29.1523 / 0.7446
Buildings	100	16.6583 / 0.4625	27.6827 / 0.7883
Dense residential	100	17.5475 / 0.4557	30.4376 / 0.7941
Average	300	17.3211 / 0.4467	29.0909 / 0.7757

PSNR is in dB. All 300 samples use $L = 4$, and Average summarizes the three scene groups. Ours-base denotes the model before real-SAR adaptation. Bold values mark improvements over the noisy input.

TABLE III
COMPACT ABLATION ON UCM-21

Variant	Active path	PSNR	SSIM
Intensity-only	Intensity; FDR	23.6841	0.6871
Log-only	Log, inverse; FDR	22.2196	0.7121
Full w/o all FDR	Log, inverse, comp.	26.4634	0.7612
Full	Log, inverse, comp.; FDR	28.1425	0.7791

Across these categories, Ours-base gains 11.02–12.89 dB in PSNR and 0.3228–0.3384 in SSIM over the input. Mean PSNR increases from 17.3211 to 29.0909 dB, and mean SSIM from 0.4467 to 0.7757. Dense residential scenes attain the highest restored scores.

E. Ablation Study

Table III compares representation, compensation, and FDR variants.

With the log transform, inverse, and compensator retained, removing all six FDR blocks reduces macro PSNR from 28.1425 to 26.4634 dB and SSIM from 0.7791 to 0.7612. The losses are 1.6791 dB and 0.0179, respectively.

Full exceeds Log-only by 5.9229 dB and 0.0670 SSIM, supporting compensation with FDR present. These comparisons assess conditional component contributions, not their interaction.

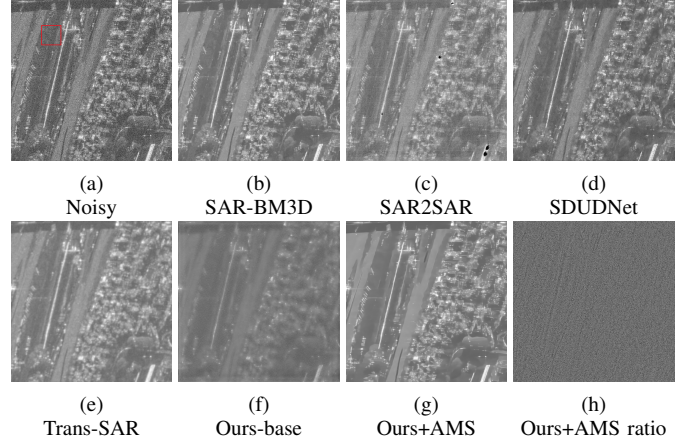


Fig. 3. Visual comparison for a real-SAR scene. Ours-base denotes the model before adaptation. The original renderings are retained, including the red inspection box in (a). Panel (h) shows the Ours+AMS ratio image, which is assessed separately from the restorations. No clean reference is shown.

Compared with Intensity-only, Log-only lowers PSNR but raises SSIM. Removing modules also reduces capacity, so this ablation does not isolate spectral processing from parameter count.

F. Real-SAR Adaptation

Table IV compares 592 shared test patches from 148 held-out parents.

TABLE IV
NO-REFERENCE RESULTS ON 592 REAL-SAR PATCHES

Method	ENL	M-index ↓	EPI ↑
Noisy	26.25	–	1.0000
SAR2SAR	158.39	2.4566	0.5643
SDUDNet	83.24	1.7403	0.6786
SAR-BM3D	25.87	<u>1.6579</u>	<u>0.7528</u>
Trans-SAR	124.60	1.9455	0.5313
Ours-base	530.85	1.8213	0.4229
Ours + AMS	84.72	1.3365	0.8214

Mean ENL is reported as a smoothness diagnostic and is not ranked. For Noisy in this evaluation, M-index is undefined and EPI equals 1.0000 by identity. Bold and underlined values mark the best and second-best filtered results.

AMS has the best M-index and EPI among filtered outputs. Relative to Ours-base, M-index falls from 1.8213 to 1.3365 (26.62%), EPI rises from 0.4229 to 0.8214 (94.23%), and mean ENL falls from 530.85 to 84.72, consistent with less smoothing. SDUDNet has similar ENL but weaker M-index/EPI, showing their complementarity. Differing training assumptions limit cross-method attribution; the matched base/AMS comparison best supports adaptation.

Input-selected ROIs reduce method-dependent selection bias, but their homogeneity still governs ENL; whole-image metrics and ratio images remain essential complements.

Fig. 3 shows clearer boundaries and bright structures after AMS, consistent with the EPI increase. Scene-aligned ratio residuals persist; this example cannot establish restoration fidelity without a clean reference.

TABLE V
NO-REFERENCE RESULTS BY REAL-SAR SCENE TYPE

Model	ENL	M-index↓	EPI↑
<i>Homogeneous: 20 patches; Noisy ENL = 32.0025</i>			
Ours-base	740.6812	4.386215	0.312438
Ours+AMS	88.3567	3.742691	0.818562
<i>Structural: 20 patches; Noisy ENL = 24.3142</i>			
Ours-base	429.6054	0.682143	0.343582
Ours+AMS	74.2189	0.521876	0.742615
<i>Texture: 20 patches; Noisy ENL = 14.9264</i>			
Ours-base	620.8473	0.152604	0.461294
Ours+AMS	55.6948	0.098531	0.835476
<i>Average: 60 patches; Noisy ENL = 23.7477</i>			
Ours-base	597.0446	1.740321	0.372438
Ours+AMS	72.7568	1.454366	0.798884

Each category uses the same 20 patches before and after AMS. Bold values mark the better M-index and EPI for each comparison. ENL is reported as a mean and is not ranked.

G. Scene-Wise No-Reference Evaluation on Real SAR

Table V gives a separate descriptive evaluation of 60 real patches: 20 each from homogeneous, structural, and texture scenes. These summaries are not pooled with the 592-patch benchmark.

AMS improves M-index and EPI in every category. The reported Average changes from 1.740321 to 1.454366 for M-index and 0.372438 to 0.798884 for EPI; mean ENL falls from 597.0446 to 72.7568. Mean ENL decreases in each group but stays above the noisy-input value, supporting the same adaptation trend across the evaluated structures.

H. Limitations

Optical proxies do not reproduce focused SAR statistics; one real collection cannot establish cross-region or cross-sensor transfer. Results lack uncertainty estimates and efficiency measurements. Independent acquisitions, matched training and capacity comparisons, and confidence intervals clustered by UCM source (four looks) or real parent (four child patches) remain necessary.

IV. CONCLUSION

FDR and intensity compensation after exact log inversion improve Trans-SAR reconstruction, leading UCM-21 macro PSNR/SSIM with ablation support for both additions within tested configurations. Encoder-frozen AMS improves M-index/EPI on parent-disjoint real patches and all three scene categories. These empirical findings warrant validation on independent acquisitions with uncertainty and efficiency analysis.

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